

Ebublio: Edge Assisted Multi-user 360-Degree Video Streaming

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ABSTRACT

Compared to traditional videos, streaming 360° videos is more difficult. We propose Ebublio, a novel intelligent edge caching framework consisting of a collaborative FoV prediction (CFP) module and a long-term tile caching optimization (LTO) module. The former integrates the features of video content, user trajectory, and other users' records for combined prediction. The latter employs the Lyapunov framework and a subgradient optimization toward the optimal caching replacement policy. Our trace-driven evaluation further demonstrates the superiority of our framework, with about 42% improvement in FoV prediction, and 36% improvement in QoE under similar traffic consumption.

Index Terms: Virtual reality—360° video—360° video streaming; Virtual reality—360° video—FoV prediction; Networking—Edge computing and caching; Optimization—Lyapunov optimization

1 INTRODUCTION

Recent years have witnessed the booming development of virtual reality (VR), which provides users an immersive and interactive experience to enjoy the virtual world. As one of the most important manifestations, 360° videos bring a new watching experience that has been widely adopted in many multimedia applications such as gaming, education, tourism, and sports. It is also a key technology that supports the development of the new paradigm *Metaverse*. Cisco Visual Networking Index (VNI) [2] indicates that 360° videos mobile data traffic will grow nearly 12-fold from 2017 to 2022.

In 360° videos, users usually need to wear a professional VR headset like HTC VIVE or simple VR-compatible mobile devices such as Google Daydream. Given the spherical features of 360° videos, a user is allowed to freely move his/her head and eye gaze to watch the most attractive portion of the video. And such a portion is called Field of View (FoV). Given the huge bandwidth overhead of transmitting the whole spherical angles of a 360° video, tile-based HTTP live streaming is widely used to balance the video transmission and user QoE [3, 6]. With this standard, video segment can be divided into many tiles with different qualities so that we can assign high quality for content in the FoV while low quality (or even blank content) for the rest of the video. However, the user's FoV can be quite dynamic and affected by many factors such as personalized watching preference and video content, making it difficult to achieve accurate FoV prediction.

Besides, providing high-bandwidth and low-latency multi-user streaming is also critical to 360° video watching. The development of 5G and mobile edge computing (MEC) provides a promising opportunity. The geo-distributed edge servers, e.g., base stations, can be used to cache the frequently requested tiles, which can significantly decrease network latency and bandwidth consumption, thus both improving the users' QoEs and saving the service cost [4, 7]. Designing such a caching system, however, is non-trivial. Given an edge node serves multiple users within the proximity, how to satisfy the users' overall QoE with fairness into account and maintain a long-term cost-effective strategy to avoid back and forth replacement is still a challenging problem.

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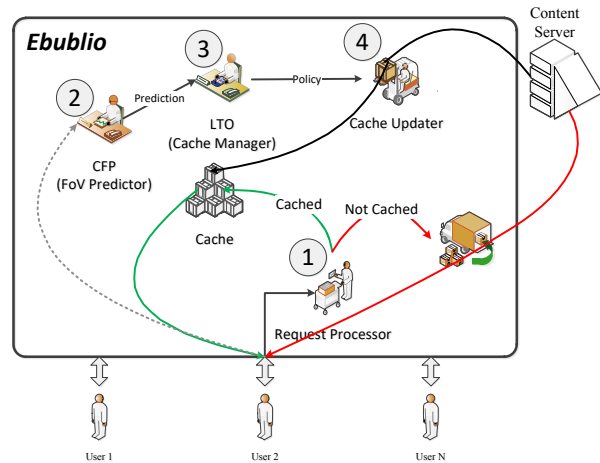


Figure 1: The Structure of Ebublio

Existing works are incapable of solving these challenges well. Most works [5] focus on a single client and adopt a non-cooperative way for video requesting, which inevitably wastes unnecessary bandwidth and computation resources from a holistic perspective. The other efforts [4] try to improve the cache hit rate for multi-users, while they are often myopic since their cache policies are only based on the current state without a long-term consideration. Furthermore, these works only care about the total user QoE, which may sacrifice those maverick users whose FoVs are largely different from the major users.

To this end, we propose Ebublio¹, a novel intelligent edge caching framework that jointly achieves user FoV prediction and 360° video tile caching to optimize the long-term user QoE and system cost. Ebublio consists of two modules: a collaborative FoV prediction (CFP) module and a long-term tile caching optimization (LTO) module. We first conduct a comprehensive spatial and temporal analysis on the user FoV trajectory and find it is highly correlated with the video content, the target user's historical trajectory, and the FoV of other users who have watched the video. Thus, we design an intelligent CFP module that integrates these three features toward collaborative FoV prediction. After the FoV prediction, the LTO module then seeks a caching policy that both maximized the overall user QoE and minimizes the system cost. It solves the long-term optimization problem with both the Lyapunov framework and dual composition as well as subgradient descent. We have conducted extensive evaluations and the real-trace driven experiments show that compared to state-of-the-art solutions Ebublio can achieve 36% improvement in QoE under similar traffic consumptions and outperforms in FoV prediction.

2 PROPOSED DESIGN

We propose Ebublio which consists of two main modules: a collaborative FoV prediction (CFP) module and a long-term tile caching

¹Ebublio is an incantation in *Harry Potter* to entrap targets(video tiles) in a bubble(edge cache).

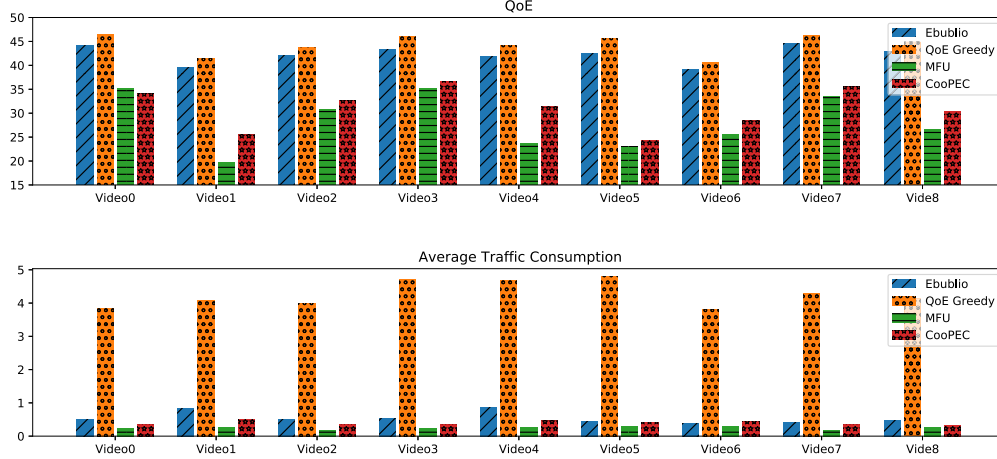


Figure 2: Comparisons of Ebublio and Baselines

optimization (LTO) module, which will be described in detail in section 2.1 and section 2.2. Fig. 1 shows the structure of Ebublio and the procedure shows as follows

1. At timeslot t , users request the next chunk, if the requested tile is in the cache, return it. If not, the edge downloads it from the content server at a lower bitrate.
2. CFP predicts future FoVs of each user and sends the results to LTO.
3. Based on the prediction, LTO calculates a cache policy.
4. Update the cache according to the policy.

2.1 CFP: Collaborative FoV Prediction

CFP predicts FoV based on multiple perspectives: historical trajectory, video content, and shared FoVs. Each component first works individually and then calculates their weighted average, where the weights are generated by Passive-Aggressive Regression model.

2.2 LTO: Long-term Tile Caching Optimization

We first leverage Lyapunov optimization to transform the long-term optimization problem into a problem in one timeslot, then solve this problem by dual decomposition and subgradient descent.

The optimization problem involves the long-term expected terms, so we seek the technique of Lyapunov optimization. Utilizing the penalty drift in Lyapunov optimization theory, the optimization problem can be written as

$$\text{objective} : \min_{S_t} \left\{ \Delta(\Theta_t) + V \times E \left[\left(-\sum_n U_{t,n} \right) + \beta B_t | \Theta_t \right] \right\} \quad (1)$$

$$s.t. 0 \leq \sum S_t \leq C \quad (2)$$

$$B_t \leq B_{max,t} \quad (3)$$

where $\Delta(\Theta_t)$ is penalty drift and $V \geq 0$ is the penalty weight.

Then we can use dual decomposition, which transforms the original problem into one master problem and several distributed subproblems, to cope with the coupling optimization problem. Then the Lagrange multiplier λ can be updated by the subgradient of the master problem.

3 EVALUATION

We use a popular dataset [8] that has nine videos watched by 48 users with an average view duration of 164 seconds.

To evaluate the performance of Ebublio, we compare it with baselines as described below:

- **QoE-Greedy:** For this method, we let Ebublio greedily optimize QoE, ignoring traffic consumption.
- **Most Frequently Used (MFU):** This method actively prefetch tiles used most frequently to replace those fewer ones.
- **CooPEC [4]:** CooPEC prefetch one tile based on FoV prediction to replace another based on previous FoV information. CooPEC also works on reducing core network bandwidth consumption and enhancing QoE for users on the edge. As far as we know, it's a state-of-the-art framework.

Fig. 2 shows the comparisons of QoE and traffic cost for different methods. From the figure, we can find that the QoE of QoE-Greedy is slightly higher than Ebublio, but the traffic cost is nearly 10 times larger. It's because QoE-Greedy updates the cache frequently and severely to achieve better QoE. Both these two methods are much better than MFU and CooPEC from the perspective of QoE. In particular, Ebublio outperforms CooPEC 36%. It's because CooPEC is designed for a single user and hasn't considered the long-term optimization problem.

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REFERENCES

- [1] L. Chopra, S. Chakraborty, A. Mondal, and S. Chakraborty. PARIMA: viewport adaptive 360-degree video streaming. In *WWW. ACM/IW3C2*, 2021.
- [2] Cisco. Cisco visual networking index: Forecast and trends, 2018.
- [3] M. Graf, C. Timmerer, and C. Müller. Towards bandwidth efficient adaptive streaming of omnidirectional video over HTTP: design, implementation, and evaluation. In *MMSys. ACM*, 2017.
- [4] A. Mahzari, A. Samiei, and R. Prakash. Coopec: Cooperative prefetching and edge caching for adaptive 360° video streaming. In *ISM. IEEE*, 2020.
- [5] F. Qian, L. Ji, B. Han, and V. Gopalakrishnan. Optimizing 360 video delivery over cellular networks. In *ATC@MobiCom. ACM*, 2016.
- [6] P. Rondao-Alface, J. Macq, and N. Verzijp. Interactive omnidirectional video delivery: A bandwidth-effective approach. *Bell Labs Tech. J.*, 16(4), 2012.
- [7] L. Sun, Y. Mao, T. Zong, Y. Liu, and Y. Wang. Flocking-based live streaming of 360-degree video. In *MMSys. ACM*, 2020.
- [8] C. Wu, Z. Tan, Z. Wang, and S. Yang. A dataset for exploring user behaviors in VR spherical video streaming. In *MMSys. ACM*, 2017.